Digital Preservation of Cultural Heritage using XR Technologies

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0 EXECUTIVE SUMMARY OF THE PROPOSAL

The "Virtual-Europe-Path" project (Digital Preservation of Cultural Heritage using XR Technologies) addresses the critical dual challenge of preserving Europe's deteriorating cultural monuments while managing the damaging physical impact of mass tourism. Current state-of-the-art digital preservation is often restricted to isolated, indoor environments and lacks the scalability required for complex outdoor archaeological sites [State of the Art]. This project advances the field by creating a pan-European, interdisciplinary framework that integrates high-fidelity 3D/4D scanning, eXtended Reality (XR), and Open Science principles to ensure the survival and accessibility of our shared heritage.
Led by Eötvös Loránd University, the consortium unites a world-class network of technological, publishing, and scientific partners:

- **Technological Innovation: Arts et Métiers (CAPLAB)** shifts advanced 4D reality capture—originally developed for industrial and healthcare sectors—into the cultural heritage domain, recording not just structures but also "craftsman intelligence" through motion and gesture capture.
- **Scientific Dissemination: Springer Nature** drives a high-impact Open Access model, transforming project data into a global "fuel" for researchers, startups, and museums.
- **Regional Expertise**: Premier institutions including Sapienza Università di Roma, NKUA (Athens), ELTE (Budapest), and Sorbonne Université provide the archaeological anchor, testing these technologies across diverse and complex scenarios—from Roman urban ruins and Greek coastal sanctuaries to Central European endangered wooden churches and Atlantic fortifications.

Key Outcomes and Innovations

- **"Virtual-first" Tourism Model**: Mitigating overtourism by providing remote, high-fidelity XR experiences that serve as "experience replicas" for fragile or restricted sites.
- **Technological Validation**: Pushing 3D/4D scanning tools to higher Technology Readiness Levels (TRL) through validation in deteriorating outdoor environments and climatic risk zones.
- **Economic and Social Impact**: Establishing a sustainable ecosystem through digital ticketing, virtual marketplaces for local artisans, and community-led narratives that preserve local identity.

By bridging the gap between scientific documentation and immersive public access, "Virtual-Europe-Path" creates a scalable, standardized workflow for the digital era, ensuring that Europe's cultural legacy is both physically protected and globally accessible.

1 TECHNICAL AND SCIENTIFIC DESCRIPTION OF THE PROJECT

1.1 THE PROJECT TOPIC AND ITS PRACTICAL RELEVANCE

Project Topic and Correlation with the Call
The project, "Digital Preservation of Cultural Heritage using XR Technologies," focuses on the high-fidelity digital reconstruction and immersive accessibility of endangered cultural monuments. By integrating 3D laser scanning (LiDAR), aerial photogrammetry, and Extended Reality (XR)—comprising Virtual, Augmented, and Mixed Reality—the proposal creates "digital twins" of historical sites. This aligns with the thematic priorities of the call by fostering

technological innovation in the preservation of European cultural identity across Mediterranean, Central-Eastern European, and Atlantic regions.

Practical Relevance
The practical relevance of this research addresses two critical global challenges: the physical degradation of heritage sites caused by environmental factors (e.g., coastal erosion, climate change) and the unsustainable pressure of mass tourism. The project offers a non-destructive preservation method, allowing for the meticulous recording of structures before further decay occurs. Furthermore, it provides a scalable solution for "Virtual-First" tourism, which unburdens fragile UNESCO sites while maintaining economic engagement through digital accessibility.

End-Products and Expected Results
The project will deliver a suite of integrated technological solutions:

- **Standardized Digital Heritage Framework:** A methodological protocol for high-precision 3D documentation in complex outdoor environments.
- **VR-based "Virtual Tour" Engine:** A high-fidelity rendering architecture (Unreal/Unity) capable of handling massive photogrammetry datasets for remote exploration.
- **AR-based "On-Site Time Machine":** A mobile module that overlays digital reconstructions onto real-world ruins in real-time for physical visitors.
- **Open Science Ecosystem:** A cloud-based repository for 3D assets and scientific documentation, ensuring long-term visibility and research integration.

Success Criteria
Success will be measured by the advancement of Technology Readiness Levels (TRL) through validation in real-world ruin scenarios. Key performance indicators include a minimum 85

1.2 PROJECT CONTRIBUTION BEYOND THE STATE OF THE ART

Recent advances in XR technologies and digital twin systems have opened new possibilities in cultural heritage preservation. However, existing solutions are often limited to isolated case studies or indoor environments.

Current limitations include:

- Lack of scalability
- Limited outdoor applicability
- Weak integration between scientific documentation and public accessibility

The project advances the state of the art by:

- Applying XR technologies in complex outdoor archaeological contexts
- Integrating large-scale, multi-region datasets
- Combining scientific documentation with immersive public access

1.3 PROJECT OBJECTIVES AND OUTCOMES

The main objectives of the project are:

- Develop high-quality 3D digital representations of cultural heritage sites
- Enable virtual access through VR/AR technologies
- Reduce physical tourism pressure on vulnerable sites
- Establish standardized methodologies for digital heritage documentation
- Foster interdisciplinary collaboration between technology and humanities
- Support open science through accessible datasets and publications

In detail, the project aims to save lesser-known or rapidly deteriorating monuments using 3D scanning and VR/AR technology, allowing them to be visited without the physical burden of mass tourism. Key targeted achievements include:

- **Technological Validation:** Testing XR platforms in complex, outdoor environments (e.g., ruins, built heritage) to increase Technology Readiness Levels (TRL) and demonstrate real-world impact.
- **Open Science and Visibility:** Generating high-quality datasets, 3D models, and VR assets as part of an open access ecosystem, maximizing visibility, citation, and future funding opportunities.
- **International Network Expansion:** Strengthening EU consortium networks by bringing together technology hubs, scientific publishers, and top-tier heritage protection partners across the Mediterranean, Central-Eastern, and Atlantic European regions.

1.4 ORIGINAL AND INNOVATIVE CONTRIBUTIONS OF THE PROJECT

The project introduces innovative contributions by combining XR technologies with scalable digital heritage workflows.
Key innovations include:

- Use of high-fidelity digital twins as reusable cultural assets
- Integration of XR environments with real-world heritage preservation strategies
- Standardized workflows for outdoor heritage digitization

Further breakthrough areas include:

- **New XR Application Domain:** Shifting advanced 4D reality capture (recording motion, gesture, sound, and physiological data) from industrial and healthcare sectors directly into cultural heritage. This represents a highly innovative use-case that aligns with EU "cultural heritage digitisation" priorities.
- **Validation in Complex Scenarios:** Pushing the technological boundaries by validating scanning technologies on outdoor, deteriorating ruins under real-world conditions, rather than controlled lab environments.
- **Open Access Revenue and Science Models:** Implementing an innovative ecosystem where digital heritage + XR research acts as fuel for an open science platform, increasing accessibility for developers, startups, and museums.

1.5 INTER-, MULTI-, OR TRANS- DISCIPLINARY CHARACTERISTICS

The project adopts an interdisciplinary approach combining computer science, archaeology, cultural heritage studies, and digital humanities. Technological development is closely integrated with heritage expertise, ensuring both scientific accuracy and user accessibility. This approach enables effective collaboration across domains and enhances the societal relevance of the project outcomes.

This interdisciplinary synergy is manifested through the following pillars:

- **Technology and Publishing:** Arts et Métiers (CAPLAB) provides cutting-edge XR tools, while Springer Nature ensures that the resulting digital heritage trend reaches the widest possible academic and public audience through open science and open access mechanisms.
- **Mediterranean Heritage Experts:** Sapienza Università di Roma and the National and Kapodistrian University of Athens bridge classical Greco-Roman archaeology, urban conservation, and climate risk assessments with modern digital documentation methods.
- **European Regional Synergy:** ELTE BTK (Hungary) brings AI-based processing and digital humanities expertise (Danube-AI), while Sorbonne Université (France) coordinates data collection on medieval fortifications and coastal erosion, ensuring a truly pan-European and cross-disciplinary framework.

2 IMPACT AND DISSEMINATION OF THE PROJECT RESULTS

2.1 DISSEMINATION AND EXPLOITATION OF THE PROJECT RESULTS

The dissemination strategy for the "Virtual-Europe-Path" project is designed to maximize the visibility, uptake, and scientific impact of the research outcomes across academic, industrial, and public sectors. By leveraging the unique strengths of our publishing partner, Springer Nature, and the technological expertise of Arts et Métiers (CAPLAB), the project ensures that its findings are accessible to the widest possible audience. **Academic and Scientific Dissemination** The primary vehicle for scientific dissemination is a robust Open Access publication strategy. This approach ensures that research remains freely accessible while maintaining high citation potential and visibility.

- **Open Access Publications**: High-quality, interdisciplinary articles will be published in journals such as Heritage Science and Open Access.
- **Open Science Ecosystem**: Beyond traditional articles, the project will release primary research data, including 3D models, scanning datasets, and VR assets, as part of an open research framework.
- **Conference Participation**: Partners such as NKUA and Sapienza will present findings at international archaeology and digital heritage conferences to strengthen their positions within EU innovation networks.
- **Thematic Cohesion**: Collaborative publications between regional partners (e.g., Sapienza and NKUA) will highlight comparative studies of Mediterranean heritage, increasing the thematic impact of the project.

Technological and Industrial Dissemination To foster innovation and market potential, the project targets developers and the tech industry.

- **Technology Validation**: The project acts as a live validation for XR technologies in complex outdoor environments, increasing the Technology Readiness Level (TRL) of the platforms used.
- **Developer and Startup Engagement**: By highlighting new XR use cases in cultural heritage, the project attracts interest from software developers and startups specializing in VR/AR applications.
- **Technology Transfer**: Arts et Métiers will utilize the project to demonstrate the practical application of 4D reality-capture, facilitating knowledge transfer to industrial and cultural partners.

Stakeholder and Public Engagement Dissemination extends to the "end-users" of cultural heritage: museums, tourism boards, and the general public.

- **Professional Networks**: Results will be shared with national and international heritage bodies, such as the Italian Ministry of Culture (MiC) and UNESCO, to inform future monument preservation strategies.
- **Virtual Heritage Exhibition**: A dedicated exhibition will be organized to showcase the VR/AR prototypes to the public, demonstrating the "Virtual-first" tourism model.
- **"Virtual-Europe-Path" Closing Conference**: A final event will gather all stakeholders to discuss the project's social impact and its role in reducing overtourism through digital alternatives.

Expoitation and Sustainability Plan The project includes a structured Exploitation Plan to ensure long-term viability beyond the initial funding period.

- **Standardization of Data**: The project will produce standardized 3D models and documentation protocols that can be adopted by other regions to create a consistent European digital heritage database.
- **Brand Positioning**: For partners like Springer Nature, the project serves to position the organization as a comprehensive knowledge platform for EU-funded research.
- **Scalability**: The methodologies developed in the Mediterranean, Atlantic, and CEE regions are designed to be adaptable to other heritage sites globally, ensuring a lasting technological legacy.

2.2 POSSIBLE APPLICATIONS WITH MARKET POTENTIAL

The market potential and exploitation strategies are deeply intertwined with the open science model and XR sector expansion:

- **Open Access Revenue Model:** The partnership with Springer Nature leverages Article Processing Charges (APCs) to ensure free access to publications, 3D models, and VR assets. This attracts not only academics but also software developers and XR startups, leading to higher download rates and increased brand reputation.
- **XR Technology Expansion:** Validating 4D capture in complex cultural heritage scenarios opens a completely new market segment for XR technologies, increasing their societal relevance and commercial viability beyond traditional industrial/health training.
- **Sustainable Tourism Economy:** By addressing the physical destruction caused by overtourism, the project provides a scalable digital alternative. "Experience-replicas" and remote access tools offer tourism boards new digital products that generate engagement while unburdening fragile UNESCO sites.

2.3 ESTIMATED IMPROVEMENTS IN THE QUALITY OF LIFE, WITH RESPECT TO CURRENT PERFORMANCE OF PRODUCTS, TECHNOLOGIES AND/OR SERVICES

The "Virtual-Europe-Path" project is designed to deliver a transformative impact across social, environmental, and economic dimensions. By integrating XR technologies with heritage science, the project addresses the critical tension between tourism-driven economic growth and the physical preservation of European history.

Environmental and Physical Heritage Impact The most immediate impact is the physical protection of "at-risk" monuments from the degradation caused by mass tourism.

- **Reduction of Physical Load**: By creating high-fidelity "experience replicas," the project allows for virtual visitation, which directly reduces the physical and aesthetic wear on fragile ruins in high-traffic areas like Rome and Athens.
- **Site-Specific Preservation**: 3D scanning and digital twins act as a permanent record for sites currently threatened by coastal erosion (Atlantic Coast) and climatic risks like storm surges or soil movement (Mediterranean).
- **Diverting Tourism Flows**: The project aims to divert tourist interest from overcrowded UNESCO sites toward lesser-known monuments, balancing the regional tourism load.

Social Impact and Quality of Life The project enhances the relationship between European citizens and their shared cultural identity.

- **Democratization of Culture**: Remote VR access allows individuals with limited physical mobility or financial constraints to "visit" and experience world-class heritage sites.
- **Preservation of "Craftsman Intelligence"**: Beyond physical structures, the project records movements, gestures, and traditional knowledge, ensuring that "craftsman intelligence" is preserved for future generations.
- **Local Identity and Co-creation**: Community engagement workshops ensure that digital representations respect and integrate local legends and identities, strengthening the social bond between residents and their heritage.

Economic Impact and Sustainability The project introduces a sustainable financial model for heritage management that does not rely solely on physical attendance.

- **New Revenue Streams**: The development of digital ticketing (B2C) and licensing frameworks for educational institutions (B2B) provides museums and local governments with diversified income.
- **Support for Local Economies**: By integrating local artisanal products and services into a "virtual marketplace," the project extends the economic benefits of tourism to local craftsmen even if physical visitors are restricted.
- **Market Relevancy for XR**: Providing a new, high-value use case for XR technology in the cultural sector increases the market and social relevance of the European tech industry.

Scientific and Technological Impact The project serves as a catalyst for innovation in the fields of Digital Humanities and XR.

- **TRL Advancement**: Testing 3D and 4D scanning in complex, outdoor, and uneven terrains (such as Greek acropolises) significantly raises the Technology Readiness Level (TRL) of current digital documentation tools.

- **Open Science Contribution**: The creation of standardized datasets and 3D assets provides the scientific community with a rich database for comparative archaeological and architectural research.
- **Interdisciplinary Synergy**: The project fosters a new research trend at the intersection of "Digital Heritage" and XR, positioning European universities at the forefront of this emerging field.

2.4 PROJECT INTEGRATION IN THE DEVELOPMENT STRATEGY OF PARTNER COMPANIES

The project is strategically aligned with the long-term development goals of the consortium partners, addressing specific institutional and market needs:

- **Springer Nature:** The project integrates directly into their transition from a traditional publisher to a comprehensive **Open Science platform provider**. By treating high-fidelity 3D models and XR assets as citable research outputs, Springer Nature addresses the market need for standardized digital heritage documentation and open access revenue models.
- **Arts et Métiers (CAPLAB / Laval Virtual):** A core strategic goal is to expand their **4D reality capture technologies** from industrial and healthcare sectors into the cultural heritage market. The project addresses their need for technological validation (increasing TRL) in complex, non-controlled outdoor environments, which is essential for their market segment expansion.
- **University Partners (ELTE, Sapienza, NKUA, Sorbonne):** The project supports their internationalization and **Digital Humanities development strategies**. It addresses the need for interdisciplinary research frameworks that combine classical archaeology with AI-based processing and climate risk assessments.

This synergy ensures that project outcomes are not only scientifically relevant but are also embedded in the sustainable business and research models of all partners beyond the funding cycle.

3 CONSORTIUM DESCRIPTION

3.1 PROJECT DIRECTOR

The coordination of the "Virtual-Europe-Path" project is led by Eötvös Loránd University (ELTE), representing the Central and Eastern European (CEE) strategic axis. As the coordinating institution, ELTE provides the administrative, legal, and academic framework necessary to manage a large-scale, multi-national EU consortium.
Dr. Krémer Aliz - Project Director and Managing Director Dr. Krémer Aliz serves as the primary coordinator and legal representative of the consortium. In this role, she oversees the alignment of all nine Work Packages (WPs) and ensures the timely delivery of project milestones.
Professional Profile and Expertise

- **Strategic Management**: Extensive experience in leading interdisciplinary research teams at the intersection of Digital Humanities and Cultural Heritage.
- **EU Project Coordination**: Specialized in the administrative and financial management of Horizon Europe and Erasmus+ frameworks, ensuring compliance with Open Science and GDPR principles.

- **Regional Expertise**: A leading advocate for the digital preservation of endangered CEE heritage, specifically focusing on the architectural vulnerability of Transylvanian wooden churches and manor houses.

Key Responsibilities within the Project

- **Consortium Leadership**: Acting as the main point of contact between the European Commission and the partner institutions (Arts et Métiers, Sorbonne, Sapienza, NKUA, ELTE, and Springer Nature).
- **Quality and Risk Oversight**: Directing the development of the Quality Assurance (QA) and Risk Management Plan to mitigate technical or logistical delays in field scanning.
- **Financial Governance**: Supervising the budget allocation and financial reporting across all partner regions (Mediterranean, Atlantic, and CEE).
- **Dissemination Strategy**: Working closely with Springer Nature to ensure that the project's high-fidelity data and scientific outputs achieve maximum global impact.

Contact Information

3.2 CONSORTIUM STRUCTURE

The "Virtual-Europe-Path" consortium is a strategically assembled group of world-class academic institutions, a leading technology center, and a global scientific publisher. This interdisciplinary team spans across Europe, covering the Mediterranean, Atlantic, and Central-Eastern regions to address the preservation of cultural heritage through eXtended Reality (XR).
Technology Pillar: Arts et Métiers (CAPLAB / Laval Virtual) As the primary technology partner, Arts et Métiers serves as a European hub for XR, Digital Twins, and 3D technologies.

- **Core Expertise**: The CAPLAB platform specializes in 4D reality capture, which includes the recording of movement, gestures, sound, and physiological data.
- **Project Role**: They provide the technical framework for "craftsman intelligence" and knowledge preservation.
- **Strategic Benefit**: The project allows them to validate their technologies in complex, real-world outdoor environments—such as archaeological ruins—thereby increasing their Technology Readiness Level (TRL).

Dissemination and Open Science Pillar: Springer Nature Springer Nature, one of the world's largest scientific publishers, acts as the bridge between research and global visibility.

- **Core Expertise**: They focus on Open Access, Open Science, and digital research ecosystems.
- **Project Role**: They manage the high volume of publications, datasets, and case studies generated by the project's interdisciplinary nature.
- **Strategic Benefit**: By utilizing an Open Access revenue model, they ensure that research reaches a wide audience including museums, startups, and developers.

Mediterranean Region: Sapienza Università di Roma and NKUA This axis represents the most densely layered and vulnerable heritage in Europe.

- **Sapienza Università di Roma**: Founded in 1303, its Department of Antiquities is a global leader in classical archaeology and monument preservation. They bring expertise in urban heritage documentation and interdisciplinary preservation theories involving law and ethics. Their involvement focuses on high-traffic sites like the Forum Romanum to test "experience replicas" as a solution to overtourism.
- **National and Kapodistrian University of Athens (NKUA)**: Greece's oldest university provides expertise in monitoring outdoor ruins, such as acropolises and coastal sanctuaries. They lead the study on coastal erosion and climatic risks, providing a critical outdoor testing environment for 3D/4D scanning technology.

Central and Eastern Europe (CEE): Eötvös Loránd University (ELTE) The CEE region focuses on endangered castles, manor houses, and wooden churches.

- **ELTE BTK**: The Institute of Archaeological Sciences is a pioneer in non-destructive research and digital archaeology in Hungary.
- **Specialized Units**: The Department of Digital Humanities and the National Laboratory for Digital Heritage lead the AI-based processing of cultural assets.
- **Project Role**: They manage the "Danube-AI" initiative, applying Machine Learning tools to heritage along the Danube.

Atlantic Coast Region: Sorbonne Université This region focuses on coastal fortifications and lighthouses threatened by erosion.

- **Sorbonne Université (Faculté des Lettres)**: Their Institute of Art and Archaeology is a center for researching medieval and early modern fortification systems.
- **Project Role**: They are responsible for leading the data collection work package, ensuring standardized protocols across all regions.

Strategic Network Expansion: University of Bologna (Unibo) The consortium remains open to integrating the University of Bologna to further strengthen the medieval heritage focus. Through their FrameLab project, Unibo offers top-tier European expertise in heritage digitization and medieval studies, adding depth to the project's historical scope.

3.3 PARTNER RESEARCH TEAM LEADERS

Dr. Hősi Rezső Rudolf (Eötvös Loránd University)

- **Age:** 42
- **Career path & Current position:** Senior Researcher at ELTE, Department of Software Technology. PhD in Computer Vision.
- **Professional experience:** 18 years in high-performance computing, 3D rendering architectures, and spatial data processing.
- **Significant achievements:**
  1. Development of the "XR-Render" framework for mobile optimization.
  2. Publication: "Efficient Point-Cloud Processing in Outdoor Environments," IEEE Transactions, 2022.
  3. Leader of the "Digital Heritage Visualization" sub-group at ELTE.
- **Distinctions:** Member of the Hungarian Academy of Sciences (MTA) Public Body.

Prof. Dr. Jean-Luc Soulié (Arts et Métiers)

- **Age:** 52
- **Career path & Current position:** Director of the CAPLAB Platform at Arts et Métiers. Expert in 4D reality capture and XR technologies.
- **Professional experience:** 25+ years in VR/AR development and technology transfer from industrial sectors to cultural heritage.
- **Significant achievements:**
  1. "Laval Virtual" Innovation Award for 4D motion capture in complex outdoor environments.
  2. Prototype: "VEP Engine" (Virtual Europe Platform) for high-fidelity 3D rendering.
  3. Standardized protocol for 4D data capture in fragile UNESCO sites.
- **Distinctions:** Member of the European XR Alliance; French National Order of Merit for Research.

Dr. Helen Zhang (Springer Nature)

- **Age:** 45
- **Career path & Current position:** VP of Open Science Strategy at Springer Nature.
- **Professional experience:** Specialist in scientific publishing, Open Access revenue models, and data-sharing ecosystems for digital assets.
- **Significant achievements:**
  1. Implementation of the "Digital Heritage + XR" Open Access publishing track.
  2. Deliverable: "Final Sustainability and Exploitation Plan" for pan-European research alliances.
  3. White Paper: "Digital Heritage Policy in the EU," presented to EU policymakers in Brussels.
- **Distinctions:** Committee Member of COPE (Committee on Publication Ethics).

Prof. Giulia Rossi (Sapienza Università di Roma)

- **Age:** 49
- **Career path & Current position:** Full Professor of Archaeology at Sapienza. Expert in Mediterranean urban conservation.
- **Professional experience:** Specialist in Greco-Roman monuments and climate risk assessments for UNESCO urban sites.
- **Significant achievements:**
  1. Development of the "Vulnerability and Tourism Impact Index" for historical centers.
  2. Co-author: "Climate Risks in Mediterranean Heritage," Nature Heritage, 2022.
  3. Lead archaeologist for the digital documentation of the Roman Forum ruins.
- **Distinctions:** UNESCO Chair on Urban Heritage Preservation.

Prof. Dr. Dimitri Papadopoulos (National and Kapodistrian University of Athens)

- **Age:** 55
- **Career path & Current position:** Full Professor of Byzantine Archaeology at NKUA.
- **Professional experience:** Specialist in long-term site monitoring and climatic risk analysis for open-air Mediterranean ruins.
- **Significant achievements:**
  1. Lead investigator for the "Climate-Heritage" EU-sensor network for ancient sites.
  2. Publication: "Erosion Patterns in Eastern Mediterranean Marble Structures," Archaeometry Today, 2021.
  3. Developer of the "Aegis-Risk" database for Byzantine fortification monitoring.
- **Distinctions:** Member of the International Council on Monuments and Sites (ICOMOS).

Dr. Claire Lefebvre (Sorbonne Université)

- **Age:** 41
- **Career path & Current position:** Senior Researcher at the Institut d'art et d'archéologie, Sorbonne Université.
- **Professional experience:** Expert in medieval and early modern architectural history, specifically focusing on Atlantic coastal fortifications.
- **Significant achievements:**
  1. Coordination of the "Atlantic Wall" digital documentation project (2018–2022).
  2. Publication: "Coastal Erosion and the Loss of Medieval Heritage," European Journal of Architecture, 2024.
  3. Development of the "Fortress-Data" documentation protocol for military heritage.
- **Distinctions:** Recipient of the Ordre des Arts et des Lettres for cultural heritage research.

3.4 PARTNER TEAM STRUCTURE

P1 – Eötvös Loránd University (ELTE)

- **Team Leader:** Dr. Hősi Rezső Rudolf
- **Key Person:** Oláh Sára (Lead Developer & UX Strategist)
  - **Role:** Oversees the "Danube-AI" integration and the user-centric design of the XR platform.
  - **Experience:** Expert in digital archaeology and AI-based heritage processing.
  - **Achievements:**
    1. UI/UX design for the National Laboratory for Digital Heritage.
    2. Coordination of the "Digital Humanities" project branch.
- **Additional Staff:** 2 AI Engineers (MI-based processing), 1 Archaeologist (Digitization expert), 2 PhD Students (Data collection).

P2 – Arts et Métiers (CAPLAB / Laval Virtual)

- **Team Leader:** Prof. Dr. Jean-Luc Soulié

- **Team Structure:**
  - 1 **4D Capture Specialist** (Knowledge and gesture preservation).
  - 2 **XR Engineers** (VR/AR platform development).
  - 2 **Field Technician** (Outdoor/complex environment scanning).

P3 – Springer Nature

- **Team Leader:** Dr. Helen Zhang
- **Team Structure:**
  - 1 **Open Access Coordinator** (APC and licensing management).
  - 1 **Data Scientist** (3D model and dataset repository).
  - 1 **Brand Manager** (Reputation and social impact).

P4 – Sapienza Università di Roma

- **Team Leader:** Prof. Giulia Rossi
- **Team Structure:**
  - 2 **Classical Archaeologists** (Roman urban history).
  - 1 **Digital Documentation Expert** (Scanning and photogrammetry).
  - 1 **LegalEthical Consultant** (Heritage theory).

P5 – National and Kapodistrian University of Athens (NKUA)

- **Team Leader:** Prof. Dr. Dimitri Papadopoulos
- **Team Structure:**
  - 1 **Environmental Monitoring Expert** (Climate risk analysis).
  - 1 **Byzantine Archaeology Specialist**.
  - 1 **GIS and Mapping Technician** (Outdoor site modeling).

P6 – Sorbonne Université

- **Team Leader:** Dr. Claire Lefebvre
- **Team Structure:**
  - 2 **Architectural Historians** (Medieval fortifications).
  - 1 **Data Collection Coordinator** (Atlantic region lead).
  - 1 **Research Assistant** (Comparative regional data).

3.5 CONSORTIUM COMPLEMENTARITIES AND SYNERGIES BETWEEN PARTNERS

The consortium demonstrates a unique interdisciplinary synergy, combining technical innovation with deep cultural heritage expertise to ensure the project's success. The collaboration is structured around three strategic pillars:

- **Technology and Publishing Synergy:** Arts et Métiers (CAPLAB) provides cutting-edge XR tools and 4D reality capture capabilities (motion, gesture, sound), which are essential for creating high-fidelity digital twins. Springer Nature ensures that these digital assets and research outcomes act as fuel for an innovative Open Science platform, reaching the widest possible academic and public audience through open access mechanisms.
- **Mediterranean Heritage Expertise:** Sapienza Università di Roma and the National and Kapodistrian University of Athens (NKUA) bridge the gap between classical Greco-Roman archaeology and modern digital documentation. Their expertise in urban conservation and climate risk assessment provides the necessary scientific grounding for the technological validation in high-stakes, outdoor environments.
- **Pan-European Regional Synergy:** ELTE BTK (Central-Eastern Europe) contributes specialized AI-based processing and digital humanities expertise through its National Laboratory for Digital Heritage. Sorbonne Université (Atlantic region) coordinates the documentation of medieval fortifications and coastal erosion.

This interdisciplinary synergy ensures that the project outcomes are scientifically accurate, technologically advanced, and socially relevant.

4 PROJECT MANAGEMENT

4.1 WORK PLAN, DELIVERABLES AND LOAD BALANCING

This section describes the distribution of tasks and the scientific programme for all partners, broken down into nine consistent work packages (WPs).

Work package list

WP No	Work package title	Work package leader	Start	End
1	Management of the project	Sapienza Università di Roma	1	24
2	Methodological Framework	ELTE BTK	1	6
3	High-Fidelity Data Acquisition	NKUA	4	12
4	XR Platform Development	Arts et Métiers	10	18
5	Overtourism Management	Sorbonne Université	13	21
6	Community Engagement	ELTE BTK	4	24
7	Economic Models	Springer Nature	19	24
8	Testing and Pilot Actions	Arts et Métiers	16	24
9	Sustainability and Diss.	Springer Nature	1	24
TOTAL Person-Months			**252 PM**	

WP1: Project Management and Coordination

WP no.	1
WP title	Project Management and Coordination
WP leader	Eötvös Loránd University (ELTE)
Involved partners	Arts et Métiers, Springer Nature, Sapienza Università di Roma, NKUA, ELTE, Sorbonne Université
Start / End month	Month 1 / Month 24
Objectives	**O1.1.** To ensure effective administrative and financial management of the project consortium. **O1.2.** To develop and implement a Quality Assurance (QA) and Risk Management plan to mitigate potential technical or logistical delays. **O1.3.** To establish a Data Management Plan (DMP) that adheres to GDPR and Open Science principles for all generated heritage data.
Description of work	**T1.1.** Administrative and financial management: Handling legal agreements, financial reporting, and communication between the consortium and the European Commission. **T1.2.** Quality and Risk Management: Development of monitoring tools and contingency plans to ensure milestones are met within the 24-month timeline. **T1.3.** Data Management and GDPR compliance: Designing protocols for the ethical handling, storage, and sharing of 3D/4D digital assets and research data.
Deliverables	**D1.1.** Final administrative and financial management report (M24). **D1.2.** QA and Risk management plan (M3). **D1.3.** Data Management Plan based on GDPR and Open Science principles (M6).

WP2: Data collection and fieldwork

WP no.	2
WP title	Data collection and fieldwork (Methodology Framework and site selection)
WP leader	Sorbonne Université
Involved partners	Sorbonne Université, Sapienza Università di Roma, NKUA, ELTE, Arts et Métiers
Start / End month	Month 1 / Month 6
Objectives	**O2.1.** To develop a vulnerability and tourism impact index to quantify physical decay and visitor pressure. **O2.2.** To conduct multi-stakeholder analyses involving local governments, academia, and residents across the Mediterranean, Atlantic, and CEE regions. **O2.3.** To finalize a site selection matrix prioritizing monuments with high deterioration and limited physical accessibility.
Description of work	**T2.1.** Development of a vulnerability and tourism impact index: Creating metrics to assess the vulnerability of sites like Roman ruins and Atlantic fortifications. **T2.2.** Multi-stakeholder analysis: Coordinating dialogue between technical partners and local heritage bodies (e.g., UNESCO, MiC) to ensure site relevance. **T2.3.** Finalization of site selection matrix: Selecting specific monuments (e.g., wooden churches, coastal sanctuaries) for high-fidelity scanning.
Deliverables	**D2.1.** Vulnerability and tourism impact index report (M6). **D2.2.** Multi-stakeholder engagement and analysis report (M8). **D2.3.** Finalized site selection matrix and fieldwork schedule (M10).

WP3: XR development and integration

WP no.	3
WP title	XR development and integration (XR Platform development and UX design)
WP leader	Arts et Métiers (CAPLAB / Laval Virtual)
Involved partners	Arts et Métiers, Sapienza Università di Roma, NKUA, ELTE, Sorbonne Université
Start / End month	Month 10 / Month 18
Objectives	**O3.1.** To develop a high-fidelity VR-based "Virtual Tour" engine using Unreal Engine or Unity. **O3.2.** To preserve "artisanal intelligence" through 4D reality capture of movements and gestures. **O3.3.** To develop an AR-based "On-site Time Machine" for real-time ruin reconstruction via mobile applications.
Description of work	**T3.1.** VR Engine Development: Building the core architecture for 3D rendering of high-fidelity datasets to create immersive virtual visits. **T3.2.** UX Research: Conducting iterative usability testing and prototype evaluations with focus groups to optimize the user experience. **T3.3.** AR Module Implementation: Creating a mobile application module that allows on-site visitors to view digital reconstructions of archaeological ruins.
Deliverables	**D3.1.** VR-based "Virtual Tour" engine prototype (M18). **D3.2.** User Experience Research and prototype testing report (M18). **D3.3.** AR-based "On-site Time Machine" mobile module (M22).

WP4: XR Platform Development and UX Design

WP no.	4
WP title	XR Platform Development and UX Design
WP leader	Arts et Métiers (CAPLAB / Laval Virtual)
Involved partners	Arts et Métiers, ELTE, Springer Nature
Start / End month	Month 10 / Month 18
Objectives	**O4.1.** Advanced XR (VR/AR) and 3D integration for heritage. **O4.2.** Preservation of "artisanal intelligence" and 4D capture. **O4.3.** Technological validation (TRL increase) in outdoor environments.
Description of work	**T4.1.** Building the core 3D rendering architecture for high-fidelity photogrammetry datasets (Unreal/Unity). **T4.2.** Iterative usability testing with diverse demographic groups to identify navigation pain points. **T4.3.** Development of an AR module for real-time ruin reconstruction via mobile application .
Deliverables	**D4.1.** VR-based "Virtual Tour" engine (M15) . **D4.2.** UX research and prototype testing report (M15). **D4.3.** AR-based "On-Site Time Machine" module (M18).

WP5: Overtourism Management and Virtual Flow Control

WP no.	5
WP title	Overtourism Management and Virtual Flow Control
WP leader	Sorbonne Université
Involved partners	Sorbonne, Sapienza, NKUA
Start / End month	Month 13 / Month 21
Objectives	**O5.1.** Address mass tourism pressure on vulnerable monuments . **O5.2.** Establish virtual alternatives and divert tourism flows . **O5.3.** Climate-sensitive digitization (coastal erosion monitoring) .
Description of work	**T5.1.** Priority strategic guidelines for digital exploration of vulnerable sites . **T5.2.** Software tools for dynamic physical ticket availability based on sensor data . **T5.3.** Communication strategy to promote sustainable VR/AR alternatives .
Deliverables	**D5.1.** "Virtual-First" tourism model (M18) . **D5.2.** Digital ticketing and capacity monitoring algorithms (M18) . **D5.3.** Awareness campaign on responsible digital tourism (M21) .

WP6: Community Engagement and Social Impact

WP no.	6
WP title	Community Engagement and Social Impact
WP leader	Eötvös Loránd Tudományegyetem (ELTE BTK)
Involved partners	ELTE, Local communities
Start / End month	Month 4 / Month 24
Objectives	**O6.1.** Foster societal connections through digital assets . **O6.2.** Involve local populations and integrate legends into narratives . **O6.3.** Measure social impact of digital heritage interventions .
Description of work	**T6.1.** Interactive sessions with residents and historians on regional digitization . **T6.2.** Enriching digital twins with authentic, crowdsourced oral histories . **T6.3.** Systematic measurement of influence on local economies and conservation attitudes .
Deliverables	**D6.1.** Community workshop series (M12) . **D6.2.** Co-creation digital archive (M12) . **D6.3.** Social Impact Assessment (SIA) report (M24) .

WP7: Economic Models and Revenue Generation

WP no.	7
WP title	Economic Models and Revenue Generation
WP leader	Springer Nature
Involved partners	Springer Nature, Local communities
Start / End month	Month 19 / Month 24
Objectives	**O7.1.** Establish financial framework for self-sufficiency . **O7.2.** Transform digital twins into economic assets for communities .
Description of work	**T7.1.** Tiered access system (Basic/Premium/Seasonal) for the VR platform . **T7.2.** Standardized licensing packages for schools and museums . **T7.3.** Digital bridge for purchasing physical local artisanal products .
Deliverables	**D7.1.** Comprehensive Business Sustainability Plan (M24) . **D7.2.** Standardized Licensing Agreement Template (M24) .

WP8: Testing, Validation and Pilot Actions

WP no.	8
WP title	Testing, Validation and Pilot Actions
WP leader	Arts et Métiers (CAPLAB / Laval Virtual)
Involved partners	Arts et Métiers, ELTE, Sapienza, NKUA, Sorbonne
Start / End month	Month 16 / Month 24
Objectives	**O8.1.** Technical and social validation of prototypes . **O8.2.** Measure resilience to environmental factors and impact on preservation .
Description of work	**T8.1.** Multi-regional pilot events (Launch Days) across four European countries . **T8.2.** Evaluation of rendering performance across diverse VR/AR hardware . **T8.3.** UX research including eye-tracking and qualitative focus groups .
Deliverables	**D8.1.** Pilot Evaluation Report (M21) . **D8.2.** Social Impact and UX Assessment Report (M24) .

WP9: Sustainability, Dissemination and Communication

WP no.	9
WP title	Sustainability, Dissemination and Communication
WP leader	Springer Nature
Involved partners	Springer Nature, All Partners
Start / End month	Month 1 / Month 24
Objectives	**O9.1.** Post-project life cycle sustainability (CHARM-EU model) . **O9.2.** Permanent legal identity and IP management . **O9.3.** Scientific dissemination of results .
Description of work	**T9.1.** Peer-reviewed publications and conference presentations (EuroVR, Digital Heritage) . **T9.2.** Flagship closing exhibition in Brussels for EU policymakers . **T9.3.** Creation of a separate legal identity (AISBL) for long-term repository management .
Deliverables	**D9.1.** Final Sustainability and Exploitation Plan (M24) . **D9.2.** Communication Toolkit and branding package (M12) .

List of Deliverables

Del. no.	Deliverable name	WP	Leader	Nat.*	Diss.**	Date
1.1	Admin/Financial Report	1	Sapienza	RE	CO	24
2.1	Vulnerability Index	2	ELTE	IS	PU	3
3.1	LiDAR/Photogrammetry Dataset	3	NKUA	DA	PU	6
4.1	VR "Virtual Tour" Engine	4	AetM	IT	PU	15
5.1	Tourism Model Strategy	5	Sorbonne	IS	PU	18
6.3	SIA Report	6	ELTE	RE	PU	24
9.1	Scientific Publications	9	Springer	IS	PU	24

**Nature: RE=Report, IS=Intellectual Service, DA=Dataset, IT=IT tool. **Diss: PU=Public, CO=Confidential.*

4.2 COORDINATION AND TASK SCHEDULE

The project timeline spans 24 months and is structured through a phased implementation of Work Packages.

A detailed Gantt chart defines task dependencies and scheduling across all WPs.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP 1: Management of the project	1.1	Administrative and financial management	Sapienza Università di Roma								
	1.2	Development of a Quality Assurance and Risk Management Plan									
	1.3	Data Management Plan (DMP) based on GDPR and Open Science principles									
WP2: Methodological Framework and Site Selection	2.1	Development of a vulnerability and tourism impact index.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	2.2	Multi-stakeholder analysis (involving local population, local governments, academia)									
	2.3	Finalization of site selection matrix (aspects: degree of deterioration, limited accessibility).									
WP3: High-Fidelity Data Acquisition and 3D Scanning	3.1	Laser scanning (LiDAR) and aerial photogrammetry (drone-based data collection).	Nemzeti Kapodisztriasz Athéni Egyetem (NKUA)								
	3.2	Texture optimization and material-specific digital preservation.									
	3.3	Metadata enrichment (integration of archaeological and art historical data).									
WP4: XR Platform Development and UX Design	4.1	Development of a VR-based "Virtual Tour" engine (Unreal Engine/Unity).	Arts et Métiers (CAPLAB / Laval Virtual)								
	4.2	User Experience (UX) research and prototype testing with focus groups.									
	4.3	Development of an AR-based "On-Site Time Machine" module (ruin reconstruction via mobile application).									

Figure 1: Project schedule (Gantt chart) of WP1, WP2, WP3 and WP4 for the 24-month period.

Work packages and deliverable due dates and global planning			Leadning Partner	Year 1				Year 2			
WP number and title	Delivery number	Deliverable		Month 1-3	Month 4-6	Month 7-9	Month 10-12	Month 1-3	Month 4-6	Month 7-9	Month 10-12
WP5: Overtourism Management and Virtual Flow Control	5.1	Development of a "Virtual-First" tourism model (remote access vs. on-site restrictions).	Sorbonne Université								
	5.2	Design of digital ticketing system and "capacity monitoring" algorithms.									
	5.3	Awareness campaign about the "responsible digital tourist" concept.									
WP6: Community Engagement and Social Impact	6.1	Community workshops on the relationship between local identity and digital representation.	Eötvös Loránd Tudományegyetem (ELTE BTK)								
	6.2	Co-creation processes: integrating local legends into the XR narrative.									
	6.3	Preparation of a social impact assessment (SIA).									
WP7: Economic Models and Revenue Generation	7.1	Digital ticketing and subscription models (B2C).	Springer Nature								
	7.2	Development of a licensing framework for educational institutions and museums (B2B).									
	7.3	Integrating local artisanal products and services into the virtual marketplace.									
WP8: Testing, Validation and Pilot Actions	8.1	Organizing pilot events in 4 countries (on-site and remote testing).	Arts et Métiers (CAPLAB / Laval Virtual)								
	8.2	Technical performance evaluation (loading speed, rendering accuracy).									
	8.3	Collect user feedback using quantitative and qualitative methods.									
WP9: Sustainability, Dissemination and Communication	9.1	Scientific publications and conference participation (Open Access).	Springer Nature								
	9.2	Organization of a closing conference and "Virtual Heritage" exhibition.									
	9.3	Development of an Exploitation Plan to adapt the technology to other regions.									

Figure 2: Project schedule (Gantt chart) of WP5, WP6, WP7, WP8 and WP9 for the 24-month period.

4.3 AVAILABLE RESEARCH INFRASTRUCTURE, AND ITS UPGRADE/DEVELOP

The project relies on a comprehensive network of specialized infrastructures provided by the consortium:

- **CAPLAB Platform (Arts et Métiers / Laval Virtual):** A European hub for XR, digital twins, and 3D technologies, providing robust infrastructure for 4D reality capture and technology transfer.
- **Digital Open Science Ecosystem (Springer Nature):** The platform infrastructure for hosting and distributing open access articles, datasets, 3D models, and VR assets.

- **Mediterranean Field Sites (Sapienza & NKUA):** Access to highly vulnerable urban UNESCO sites in Rome and weather-exposed open-air ruins in Greece, serving as the physical infrastructure for technological testing and validation.
- **Digital Humanities Network (ELTE BTK):** The National Laboratory for Digital Heritage and the Danube-AI initiative in Hungary, supplying the computational and methodological infrastructure for AI-driven heritage processing.
- **Data Collection Network (Sorbonne Université):** Established infrastructure for coordinating architectural and historical documentation, particularly concerning fortifications and coastal heritage.

4.4 MANPOWER ALLOCATION

List of research team members		Position in the project	PM**	Cost (RON)
ELTE - Coordinator	Dr. Krémer Aliz	Project Director	18	120,000
	[Vacant]	Senior Software Engineer	12	80,000
	[Vacant]	Financial/Admin Manager	12	60,000
	[Vacant]	PhD Researchers (2)	24	90,000
ELTE - Partner	Dr. Hősi Rezső Rudolf	P1 Responsible	12	70,000
	Oláh Sára	Key Researcher (UX)	10	60,000
	[Vacant]	AI Engineers (2)	16	90,000
	[Vacant]	PhD Researchers (2)	12	40,000
Arts et Métiers (AetM)	Prof. Dr. Jean-Luc Soulié	P2 Responsible	10	50,000
	[Vacant]	XR Engineers (2)	20	40,000*
	[Vacant]	4D Capture Specialist	12	30,000*
Springer Nature (Springer)	Dr. Helen Zhang	P3 Responsible	6	25,000
	[Vacant]	Open Access Coordinator	12	15,000*
Sapienza (Sapienza)	Prof. Giulia Rossi	P4 Responsible	12	40,000
	[Vacant]	Archaeologists (2)	18	20,000*
NKUA	Prof. Dr. Dimitri Papadopoulos	P5 Responsible	12	40,000
Sorbonne Université	Dr. Claire Lefebvre	P6 Responsible	12	40,000
TOTAL			**252**	**800,000**

*Own budget contribution (Partner share).
*To be indicated with respect to the total project duration (24 months).
Indicate also the non-permanent staff (vacant positions) to hire – doctoral and post-doctoral researchers, interns, etc.

4.5 PROJECT BUDGET AND PARTNERS SHARE

The total estimated budget for the **"Virtual-Europe-Path"** project is **€4,000,000**, distributed over a **24-month** duration. This budget is strategically allocated to support high-fidelity data acquisition, cutting-edge XR development, and a robust Open Science dissemination framework.

Budget Breakdown by Work Package (WP) The allocation reflects the high costs of technical infrastructure in the scanning phase and the development of the VR/AR engine.

Table 1: Budget Breakdown by Work Package (WP)

WP #	**Work Package Title**	**Budget (€)**	**Primary Cost Drivers**
WP1	Management of the project	280,000	Admin staff at ELTE, QA, and Risk management.
WP2	Methodology & Site Selection	320,000	Stakeholder analysis and vulnerability indexing.
WP3	Data Acquisition & 3D Scanning	1,000,000	LiDAR equipment, drone operations in complex terrain.
WP4	XR Platform Development	800,000	Software engineering and 4D reality capture.
WP5	Overtourism Management	320,000	Capacity monitoring algorithms.
WP6	Community Engagement	320,000	Field workshops and social impact assessment.
WP7	Economic Models & Revenue	280,000	Licensing frameworks and virtual marketplace.
WP8	Testing & Validation	400,000	Pilot events and technical validation.
WP9	Dissemination & Communication	280,000	Open Access APC fees and final exhibition.
Total		**4,000,000**	

Partner Budget Allocation (Estimated Shares) Budget shares are distributed based on the technical intensity of tasks and regional leadership roles.

- **Eötvös Loránd University (ELTE) - Coordinator (€550,000)**: Focuses on overall project management, administrative costs, and coordinating CEE site selection.
- **Arts et Métiers (CAPLAB) (€850,000)**: Receives the largest share for XR platform development, 4D reality capture, and technology validation.
- **Springer Nature (€400,000)**: Allocated funds for managing the Open Access ecosystem, publication fees (APC), and data platform value enhancement.
- **Sapienza (Italy) (€550,000)**: Field scanning of urban ruins and overtourism social mission.
- **NKUA (Greece) (€550,000)**: Outdoor scanning in complex terrains and climate risk documentation.
- **ELTE (Hungary) (€550,000)**: AI-based processing and "Danube-AI" digital humanities integration.
- **Sorbonne Université (€550,000)**: Coordination of high-fidelity data collection and Atlantic erosion studies.

Table 2: Detailed Cost Categories

Category	**Estimated (€)**	**Justification**
Personnel	2,600,000	Researchers, XR engineers, and administrative coordinators across 4 regions.
Travel & Subsistence	300,000	Fieldwork for scanning at remote sites and international consortium meetings.
Equipment	500,000	LiDAR, high-performance drones, and VR/AR hardware for pilots.
Other Goods & Services	400,000	Open Access publication fees (APC), licenses, and final exhibition costs.
Indirect Costs (25%)	200,000	Standard EU overhead for institutional infrastructure.
Total	**4,000,000**	

Detailed Cost Categories

Budgetary Synergy and Efficiency The budget utilizes Springer Nature's Open Access model to ensure that publication costs (APC) result in higher citation rates and broader societal impact, justifying the dissemination spend. Technological costs are optimized by using Arts et Métiers' existing CAPLAB infrastructure, which is upgraded through the project rather than built from scratch.

5 REFERENCES